

ANIRUDDHO BISWAS

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Address

Present: Dhaka, Bangladesh

Permanent: Phultala, Khulna

EDUCATION

Bachelor of Science in Mechanical Engineering

June 2026

Bangladesh University of Engineering and Technology (BUET)

CGPA: 2.85 / 4.00

Notre Dame college, Dhaka(HSC)

2020

WORK EXPERIENCE

Design Manufacturing Workstation

[WeMake](#)

- Co-ran a custom 3D printing and product design business during undergraduate study on a single Creality CR10 (FDM, $300 \times 300 \times 400$ mm); handled 4–5 orders per month spanning decorative lamps and functional mechanical parts for student project requirements.
- Executed the full design-to-delivery pipeline: geometry modelling in Blender and SolidWorks (with structural/multiphysics simulation pre-print in Ansys to validate fit and load paths), slicing in OrcaSlicer with material-specific parameter tuning, and post-processing via sanding and painting; ran PLA+ as primary material with selective PETG for higher-temperature or strength-critical parts.

DhumketuX-Intern

2025-2026

- Worked with embedded systems implementations of reliability engineering, and fault tolerance system designing theories.
- Testing of built prototypes for flight related computations reliability, and recovery modes.

TECHNICAL SKILLS

- **Mechanical Design (CAD/CAM):** SolidWorks, Autodesk Fusion 360 (generative design), AutoCAD. Product designing within market constraints.
- **Simulation & Analysis:** ANSYS, SolidWorks: parametric simulations, scripting, batch simulations.
- **Embedded & IoT Systems:** ESP32, Arduino, C/C++, FreeRTOS; designed and validated IoT-integrated hardware for real-time monitoring and fault detection.
- **Fabrication:** Prototype fabrication from CAD (3D printing, Creality CR10); sanding and surface finishing.
- **Programming:** Python, C, C++; data analysis, automation scripting, AI-assisted design workflows (Claude Code / Codex integration, custom agent development), LangChain, LangGraph custom agentic orchestrations, RAG.
- **Software:** MATLAB, Blender, OrcaSlicer, Microsoft Office Suite.

PROJECT GRANTS

ASHRAE UG Equipment Grant

April 2025

- Project: Spiral Plate Heat Exchanger — secured for R&D and prototyping of heat transfer equipment design.

RELEVANT PROJECTS

Spiral Plate Heat Exchanger — ASHRAE UG Equipment Grant

[ASHRAE](#)

- Designed and fabricated a Spiral Plate Heat Exchanger prototype: geometry modeled in SolidWorks, thermal-fluid performance characterized in ANSYS Fluent, and physical prototype built and experimentally validated.
- Awarded ASHRAE Undergraduate Equipment Grant for demonstrated R&D merit; project selected for the ASHRAE Grant Showcase.

Mechanical Design Exploration & Optimisation Framework

[github](#)

- Built an end-to-end pipeline for a Macro Scale accelerometer flexure: scripted 1,296 ANSYS Workbench FEA runs via **Ansys Scripting** and dataset generation over a 4-parameter design space (beam height, length, fillet, width), trained a 20-model MLP ensemble predicting 6 structural outputs (modal frequencies 1–4, max deformation, max stress), achieving $R^2 \geq 0.969$ and $\text{MAPE} \leq 4.00\%$ across all

targets, replacing minute scale FEA runs with millisecond surrogate inference.

- Computed exact Jacobian sensitivities ($\partial \text{output} / \partial \text{input}$) across all 1,296 design points by per output — producing global mean absolute sensitivity heatmaps and input-space colormaps that identify the dominant geometric drivers for each structural response.
- Ran NSGA-II multi-objective optimisation targeting simultaneous minimisation of f_1 and maximisation of f_2 for SDOF cross-axis modal separation; extracted a 100-solution Pareto front spanning $f_1 \in [100, 244]$ Hz and $f_2 \in [747, 1443]$ Hz, and validated output manifold structure via PCA on 5,000 random predictions.

Fault Tolerance Capable Firmware Management System

[github](#)

- Designed and implemented a fault-tolerant firmware framework on ESP32 targeting self-healing flight controller applications; Architected with OTA firmware management and in flight flash and RAM re-writings to patch faults.
- Implemented two-layer software TMR to run computations in this firmware management system and avoid using normal c++ operations solely. Resulting in Temporal fault tolerance, TMR in one place for mathematical operations.
- Validated under hardware fault injection using `esptool`: each OTA partition corrupted in turn, with the firmware detecting invalid firmware partitions and autonomously repairing from a valid peer partition, and resuming concurrent TMR sensor processing without interruption — all three scenarios confirmed PASS on physical hardware.

Agentic AI Assistant — Momobot

[github](#)

- Built a local-first agentic AI system using LangGraph and Ollama, featuring a ReAct reasoning loop with a structured multi-tool ecosystem spanning file I/O, persistent shell execution, web search, vision/OCR, and nested subagent delegation for context isolation and cleaning.
- Designed a persistent task-planning layer with dependency-aware step sequencing, failure propagation, and mid-session replanning — enabling the agent to adapt to unexpected outcomes without restarting, analogous to a self-correcting automated workflow.
- Implemented a rolling context compaction system that summarises older session history when token usage exceeds a configurable threshold, maintaining coherent long-horizon reasoning within fixed LLM context window constraints.
- Implemented a memory system with management tools for permanent user-context storage, retrieval, and in-session modification.
- Featuring slash command setups for definitive functions, toggling reasoning mode, and permanent reasoning configuration for the subagent architecture for structured delegation and result tracking.